Experiment:-3

Problem Statement:-

Develop a network simulator using Cisco Packet Tracer to analyze packet delay, loss, and end-to-end throughput. Implement various routing algorithms and measure their impact on network performance under different traffic conditions.

Aim:- **Network Simulator for Analyzing Packet Delay, Loss, and Throughput**

Procedure:-

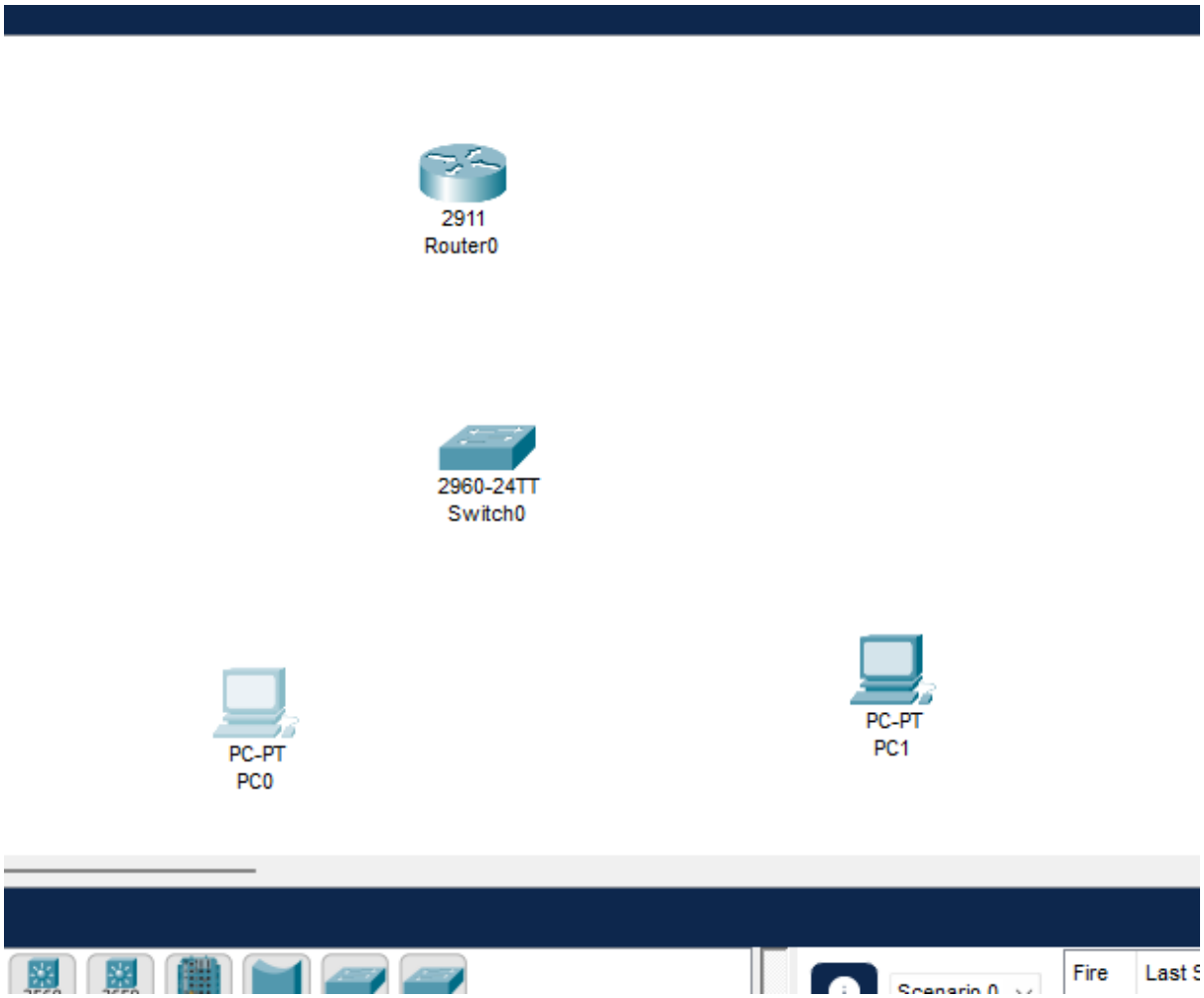
PHASE 1: Build the Network

Step 1: Open Packet Tracer

- Click **File → New**

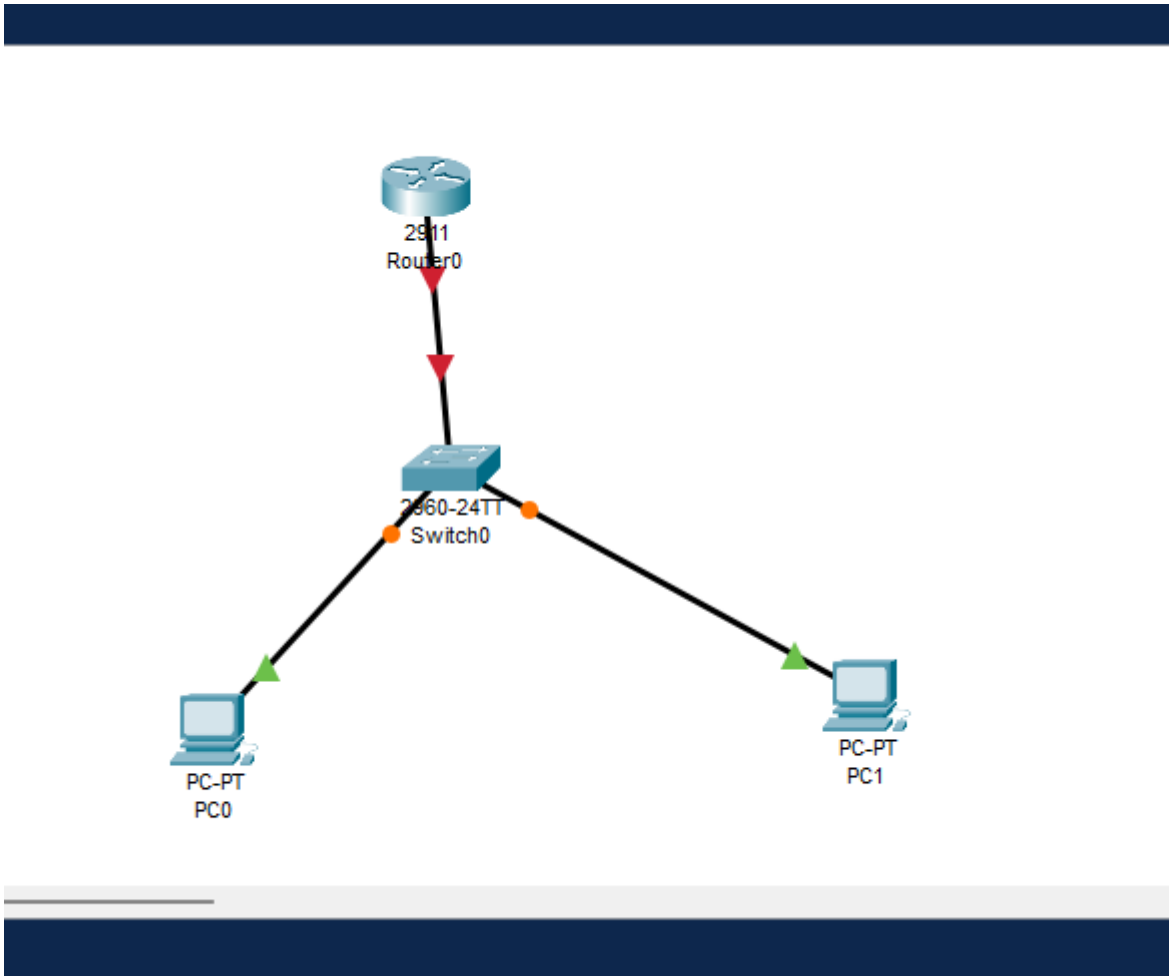
Step 2: Add Devices to Workspace

Drag and drop devices:-



Step 3: Connect Devices with Cables

Use **Copper Straight-Through** cable:



PHASE 2: Configure Router

Step 4: Click Router → CLI tab

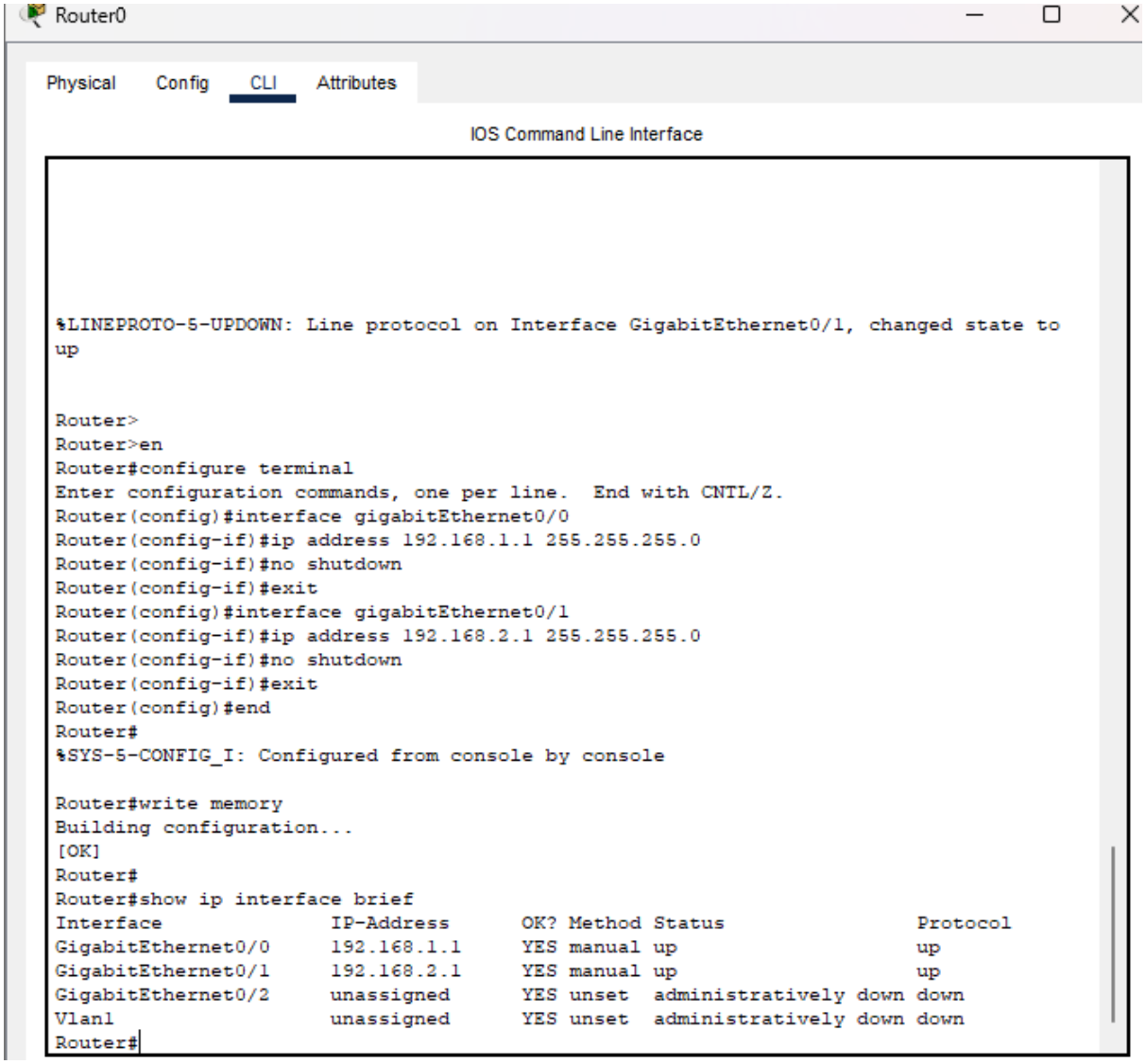
Physical Config CLI Attributes

IOS Command Line Interface

```
Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>en
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface gigabitEthernet0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#
```

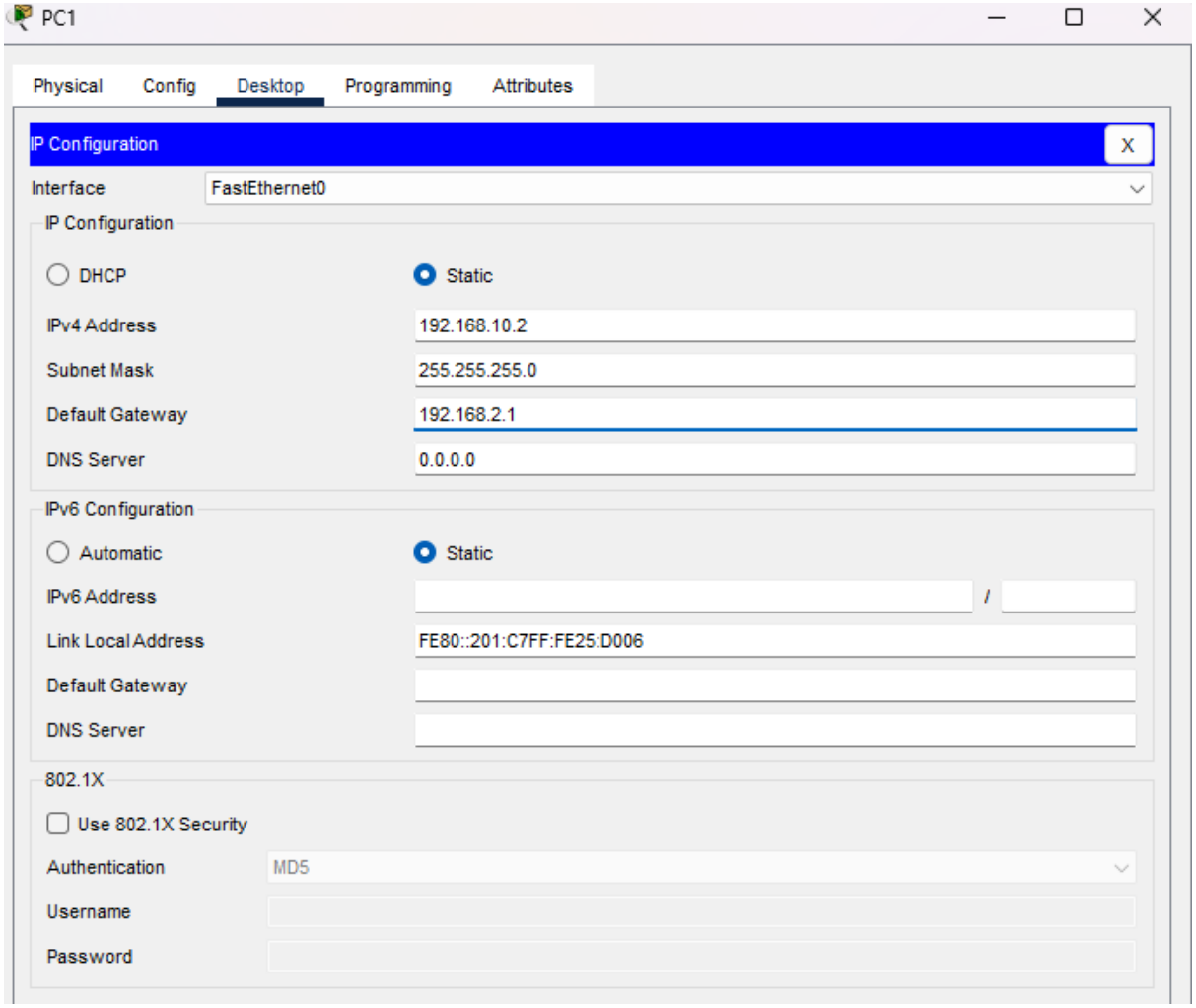
Copy Paste

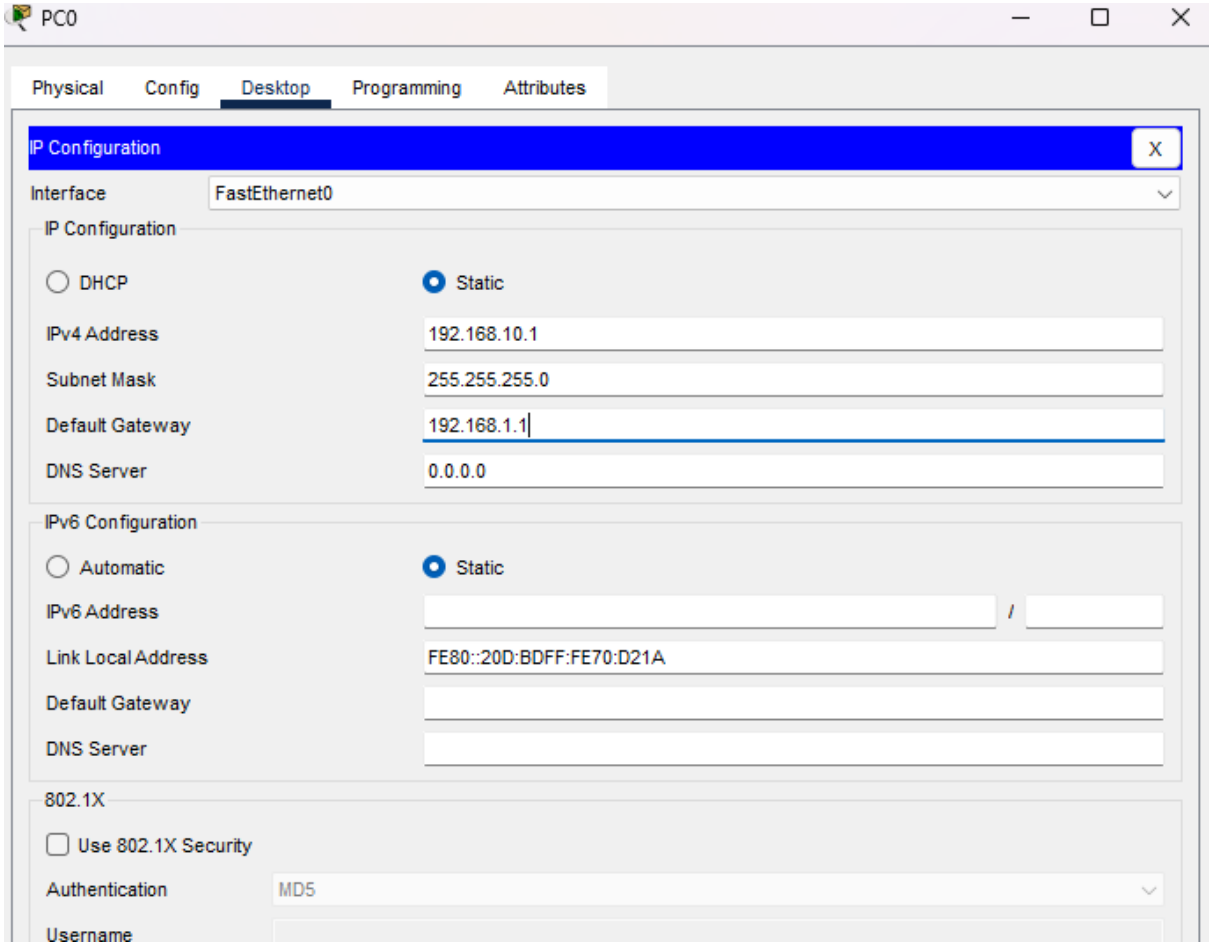
Step 5: Verify Router



PHASE 3: Configure PCs

Step 6: Click each PC → Desktop → IP Configuration


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.10.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::201:C7FF:FE25:D006
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password



PHASE 4: Static Routing

Step 7: Add Static Routes on Router

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     192.168.2.1     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 192.168.1.0 255.255.255.0 gigabitEthernet0/0
%Default route without gateway, if not a point-to-point interface, may impact
performance
Router(config)#ip route 192.168.2.0 255.255.255.0 gigabitEthernet0/1
%Default route without gateway, if not a point-to-point interface, may impact
performance
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#
```

PHASE 5: Test Connectivity

Step 8: Ping from PC0

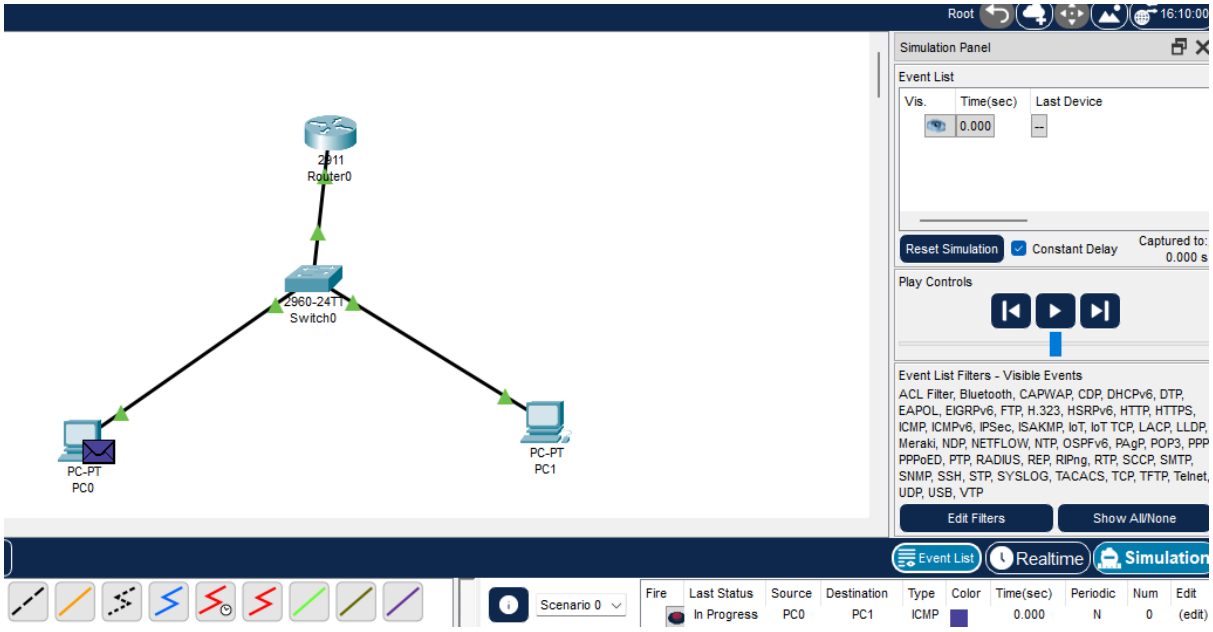
- Click **PC0 → Desktop → Command Prompt**

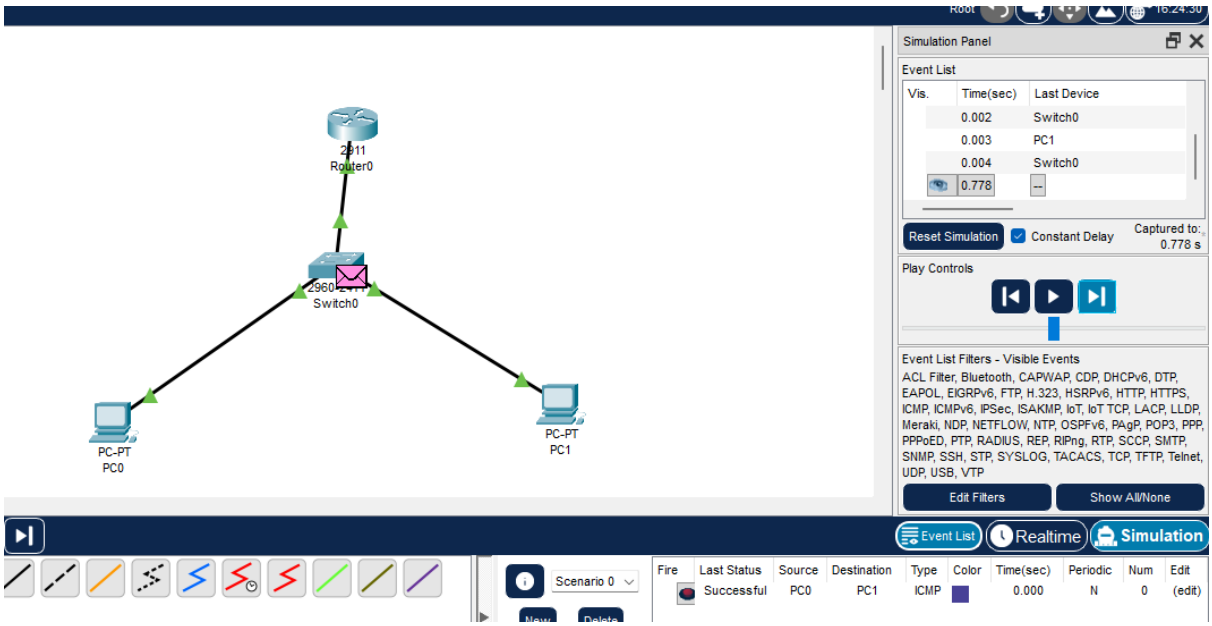
```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=2ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

PHASE 6: Simulate Traffic

Step 9: Use Add Simple PDU

- Click the **envelope icon** (Add Simple PDU) in the toolbar
- Click **PC0** then click **PC1**
- A packet will appear on the workspace



Step 10: Switch to Simulation Mode

- Click **Simulation** tab (bottom right)
- Press **Play** or **Step** button
- Watch packets travel PC0 → Switch0 → Router → Switch1 → PC1